

Aadarsha Gopala Reddy

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SUMMARY

ML engineer who built end-to-end pipelines processing 790,000 naturalistic driving trips and trained recommendation systems serving seven user personas, seeking data/ML engineering roles.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, C++, R.
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, React, Node.js, FastAPI, Socket.IO, Streamlit.
Tools & Technologies: L^AT_EX, Git, AWS, Docker, PostgreSQL, MongoDB, Snowflake, Apache Airflow, Spark, Gemini/OpenAI APIs, Microsoft 365, Google Workspace.
Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.

PROFESSIONAL EXPERIENCE

Washington University in St. Louis St. Louis, Missouri, USA
Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale **January 2026 - May 2026**

- Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing, and query optimization.
- Designed oral midterms as mock technical interviews, requiring students to reason through pipeline architecture decisions and troubleshoot system failures under real-world constraints.

Master's Thesis Research **August 2025 - May 2026**

- Built an end-to-end pipeline processing 790,000 naturalistic driving trips from 304 older adults, engineering vehicle-agnostic weekly behavioral features to predict cognitive impairment status.
- Benchmarked six models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-participant-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).

Crittero, Inc. Remote
AI Engineer Intern **June 2025 - August 2025**

- Generated 100,000 labeled social media posts across seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Trained a TensorFlow/Keras recommendation model achieving 80% accuracy (30-percentage-point gain over random baseline) with persona-type, content-category, and temporal embeddings, plus a collaborative-filtering cold-start fallback for new users.

Lab714 Boca Raton, Florida, USA
Data Analyst Intern **June 2023 - May 2024**

- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.
- Analyzed IoT environmental sensor data on AWS, building client-facing dashboards that surfaced chemical imbalances and equipment alerts, helping clients cut pool maintenance costs by 20%.

PROJECTS

Washington University in St. Louis St. Louis, Missouri, USA
Red-Blue Visual Auto Defender **August 2025 - December 2025**

- Built an automated red-blue teaming pipeline for VLM security, generating visual jailbreak attack images and auto-generating OCR-based Python defenses validated with accuracy, precision, and recall metrics.

EDUCATION

Washington University in St. Louis St. Louis, Missouri, USA
M.S. in Computer Science **August 2024 - May 2026**

- *Thesis:* Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data

Ohio Wesleyan University Delaware, Ohio, USA
B.A. in Computer Science and Data Analytics **August 2020 - May 2023**

- *Minor:* Economics | *Honors:* Golden Bishop Award, Mortar Board, Florence Leas Prize, Dean's List

LEADERSHIP

Washington University in St. Louis St. Louis, Missouri, USA
Vice President of the Graduate-Professional Council (GPC) Chamber, GPSC **May 2025 - May 2026**

- Chaired the Constitution Committee to draft and ratify founding governing documents; built committee rosters via ranked-choice preference allocation; represented graduate students to university leadership.